Karina Esquivel
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https://karinaamores.github.io/Karina-Portfolio-Website/index.html

Professional Objective

Transportation Analytics and Business Intelligence professional with experience transforming complex infrastructure, asset management, and operational data into actionable insights using Power BI, SQL, Python, ArcGIS, and Oracle databases.

Supporting transportation planning and performance management initiatives at Missouri Department of Transportation, contributing to analytics and reporting efforts associated with a $14B transportation portfolio. Proven track record of developing executive dashboards, optimizing reporting processes, improving data quality, and delivering data-driven solutions that enhance decision-making.

Recognized as a two-time finalist in MoDOT's Innovation Challenge for developing Power BI solutions that improved visibility into bridge conditions and asset management performance.

My objective is to apply my skills in analytics, automation, and user-focused design to turn complex data into clear insights. I have built scheduling solutions in Python that reduced manual input and developed dashboards that have guided decision-makers. My goal is to contribute to smarter systems, continue my growth, and improve my strategy that transforms raw data into informed, intuitive discoveries.

Certificates

- Learning Hubspot CRM - LinkedIn
- Excel PivotTables: Mastering PivotTables and Pivot Charts - LinkedIn
- Transportation Planning Process - National Transit Institute
- Learning Power BI Desktop - LinkedIn
- Power BI Essential Training - LinkedIn
- Getting Started with Power BI - LinkedIn
- Data Analytics Course & Internship - iXperience
- Learn SQL Course - Codecademy

Areas of Expertise

Power BI	Project Coordination	Reporting	Data Curation
Google API	SQL	Forecasting	Data Modelling
MS Office 365	C#	KPI Analysis	Data Visualization
CSS / HTML	ETL	Machine Learning	Spanish
Python	Customer Support	AutoCAD	Korean

Education

iXperience Boot Camp - Data Analytics (Remote)
Completed: June 2022

- Completed a rigorous data analytics bootcamp focused on data wrangling, hypothesis testing, data dashboarding, Python, SQL, ML, and data visualization with Tableau, Plotly, and Seaborn

Bachelor of Arts in Mathematical Sciences — Westminster College (Fulton, MO)
Minor: Pre-Engineering — Graduated: May 2022

Associate of Science in Engineering with Honors — State Fair Community College (Sedalia, MO)
Graduated: May 2019

Independent Study & Cultural Immersion (South Korea) — August 2019 - June 2020
July 2023 - February 2024

- Pursued immersive culture and language experiences while independently enhancing technical skills, including coding bootcamp study and practical Python development. Developed adaptability, cross-cultural communication, and resilience—qualities that have strengthened my professional and analytical approach.

Experience

Transportation Planner
MoDOT (Jefferson City, MO - Hybrid) — July 2024 - Present

- Pioneer and founding member of the new Transportation Planning Analytics Group, leading innovation initiatives
- Lead Asset Management, administering federal funds and monitoring allocations within a $14B budget to ensure compliance, efficiency, and long-term sustainability
- Liaison between MoDOT and local agencies, regional planning commissions, MPOs, and the public to align statewide priorities and regional needs
- Coordinate cross-functional schedules, mitigate delays, and execute corrective actions to keep multimillion-dollar projects on track
- Created seven regional presentations derived from a statewide infrastructure report, localizing insights and facilitating district-level planning discussions
- Identified patterns in asset allocation and spending to improve statewide forecasting performance as rural areas were not getting as much attention
- Identified a $1,000 reporting discrepancy in district asset totals, triggering a fix that improved data accuracy and statewide tier compliance
- Overseeing the organization and development of the Department's Performance Tracker (DTracker) and interactive dashboards, utilizing Everviz and Drupal to create live, public-facing charts on MoDOT's website, managing quarterly metrics and visual analytics for both internal and external reporting
- Monitor quarterly performance indicators for MoDOT's DTracker, enhancing visibility into future infrastructure needs
- Develop and launch surveys for external partners, generating informative data used in MoDOT's biannual performance reporting enabling clearer performance metrics and more responsive planning initiatives
- 2x consecutive finalist for MoDOT's Innovation Challenge for Power BI dashboard innovation, recognized for transforming bridge and asset management data into

conclusive decision-making tools for what needs improvements and which districts are behind/need help
- Spearheaded the design and implementation of an award-winning Bridge Conditions Dashboard and Asset Management Plan Dashboard, reducing time-to-insight and enabling interactive district-level, period-year, road-type filtering
- Recognized department expert on integrating ArcGIS with Power BI; developed dynamic mapping layers using ArcGIS Online for data storytelling and decision support
- Developed and optimized custom SQL queries for data extraction, transformation, and loading (ETL) from Oracle (TMS), supporting BI dashboard creation and improving data accessibility
- Revamped the TMS Pavement Tool with slicers, pivot tables, and added contextual columns, optimizing it for stakeholder clarity and filtering
- Continuously improving ArcGIS integration within Power BI to better visualize infrastructure data and enhance spatial comprehension across internal teams
- Worked with high-volume transportation datasets, incorporating elements of Big Data processing and geospatial analytics for performance monitoring

Python Developer
Freelance Work (Remote) February 2024 - Present
- Built an advanced Swiss-style scheduling system using Python, tailored for the ECNL soccer league, automating match generation while factoring in ranking, geography, and prior match history
- Developed multiple custom pairing algorithms including round-robin and random-conference-balanced matching, enabling diverse tournament structures and scheduling logic
- Enhanced scheduling accuracy using data-driven forecasting models to simulate tournament outcomes and optimize fairness across matchups
- Utilized Python libraries such as:
    - pandas – for manipulating DataFrames and processing tabular data
    - collections
      Counter – for conference frequency analysis and balancing
    - itertools
      chain – to streamline nested group iteration
- Applied data modelling and object-oriented programming (OOP) to model players, match histories, and scheduling logic via reusable class structures (Player, schedule_match, etc.)
- Engineered recursive and rotating logic to automate team placement while ensuring:
    - No duplicate matches
    - Balanced regional representation
    - Rating fairness
- Wrote complex grouping algorithms handling uneven team distributions with fallback logic for groups of 4, 6, or flexible sizes based on league needs
- Implemented match history tracking using dictionaries and sets to prevent repeat pairings across multiple scheduling rounds
- Executed data validation, conflict resolution, and edge-case handling to ensure fair outputs under tight logical constraints

- Performed data curation and preparation to ensure clean, consistent datasets for analysis and dashboarding
- Collaborated directly with non-technical stakeholders to implement and validate automated scheduling outputs in Google Sheets environments
- Kept code clean and modular, for easy understanding, updatable, and expandable for future development
- Designed and deployed a custom analytics dashboard inside Google Sheets as a new ribbon option using Google Apps Script, HTML/CSS, and Google Charts/ECharts
- Automated KPI calculations (average rank difference, total games, highest ranked conference) directly from live Google Sheets data
- Created interactive dashboards - a main and a quick side pop-up dashboard, with dropdown filters, color-coded visualizations, and real-time data refresh
- Integrated a hidden color-mapping system for conference themes and automated event recognition
- Delivered documentation and end-user guidance for non-technical users, ensuring smooth adoption and long-term maintainability

Data Entry and Mapping Analyst
Jensen Partners (Remote) October 2022 - August 2023

- Crafting executive-gradel presentations and advancing critical business development efforts
- Led quarterly updates to hundreds of alt-investment org charts using mixed research methods, fueling the firm's core analog PowerPoint system relied on by clients
- Executed a high-visibility visualization pipeline supporting executive reporting and client deliverables
- Tracked and forecasted quarterly organizational changes to inform leadership presentations and business development strategy
- Supported the implementation of a high-volume visualization workflow that delivered real-time organizational structure to clients
- Served as the primary liaison between research and digital teams, ensuring data accuracy, design standards, and timely execution of deliverables
- Improved data accuracy and data quality assurance by 90% through the development and enforcement of rigorous validation protocols and cross-functional collaboration, saving countless hours of rework and building trust across teams
- Brought energy and attention to detail into daily deliverables, team retrospectives, and dashboard improvements, raising both output data quality and internal morale
- Coached and supported team members on data visualization and cross-functional communication best practices, fostering a culture of clarity, precision, and continuous improvement

Data Analyst Intern
AMNESIA MEDIA (Remote) June 2022 - July 2022

- Generated a liability analysis report based on internal research, providing insights and recommendations that informed executive decision-making

- Partnered with data scientists to validate forecasting and reporting accuracy of AI Risk Analysis model (80% reliability) boosting team confidence in the model's reliability within just one month
- Utilized Python for data wrangling and transformation of 2,000+ data points of images, text and video; applied Seaborn and Plotly for statistical visualizations transforming raw data into compelling, decision-ready narratives
- Developed interactive dashboards in Tableau to present data findings, improving stakeholder understanding of risk categories
- Produced trend-based liability projections for executive decision-making
- Created a final Google Slides presentation that combined visual storytelling and analytics, making technical discoveries understandable to non-technical audiences
- Integrated live coursework from the iXperience Analytics Bootcamp to real-world projects, enhancing practical skills in data interpretation, visualization, and statistical thinking

Design/Research Developer
DYB-CREO (Seoul, South Korea) March 2020 - June 2020

- Co-authored and published two K–5 English coursebooks, designing age-appropriate language exercises, lessons, and quizzes sharpening skills in content development, project management, and deadline execution
- Conducted biweekly education research projects to improve lesson design, leveraging Google Workspace for collaboration and content design
- Created engaging PowerPoint visuals and interactive content daily, improving student comprehension and engagement
- Transcribed audio lesson content into clear, standardized study documents, enabling consistent instruction across classrooms and simplifying future curriculum updates
- Created assessments and instructional materials aligned with learning standards, enabling more effective teaching and performance tracking
- Collaborated cross-functionally with curriculum leads and educators to turn abstract concepts into practical, accessible materials

Hobbies

- Renovating my house
- Player for the WPSL Missouri Reign Division I Semi-Pro Women's Soccer Team
- Travel
- Cooking
- Volunteer blood drive